# Joshua Placidi

JoshuaPlacidi@gmail.com — Website — LinkedIn — GitHub — Google Scholar

## Education

**PhD in AI and Machine Learning** — **Imperial College London**
April 2025 – Present

- Researching scalable biomedical foundation models, supervised by Aldo Faisal and Marek Rei.
- Member of Nightingale AI, a UK sovereign AI project developing clinical foundation models.
- Investigate multimodal fusion, EHR foundation models, NLP, and world models.
- Run distributed training and evaluation on GPU clusters including Isambard-AI.
- Organise Imperial's Natural Language Processing Reading Group (NLP-RG).

**MSc Artificial Intelligence** — **University of Edinburgh**
September 2021 – September 2022

- Grade: Distinction
- Dissertation: Feature and Multi-Label Propagation in Graph Neural Networks [code].
- Supervised by John Pate, Tania Bakhos, and Pawel Pomorski, all machine learning research scientists at Amazon at the time.

**BSc Computer Science** — **Swansea University**
September 2017 – June 2020

- Grade: First Class Honours
- Dissertation: Fantasy Premier League bot scoring 20% above the average human player [code].
- Supervised by Daniel Archambault.

## Publications

**Sequence Composition in EHR Foundation Model Pretraining** — Preprint (under review) – 2026 [arXiv]
**J. Placidi**, I. Stibbe, A. J. Balston, M. Rei, and A. A. Faisal

**Block Diffusion for Longitudinal EHR: Accuracy, Efficiency, and Representation Trade-offs** — WMHS Workshop @ NeurIPS 2026 [arXiv]
A. Gupta, **J. Placidi**, B. C. Delaney, and A. Russo

**Constructing and Validating Memorisation Audits for EHR Foundation Models** — InfPriv Workshop @ NeurIPS 2026 *Camera-ready forthcoming*
I. Stibbe, **J. Placidi**, P. Ramnarayan, and A. A. Faisal

**Pretraining EHR Foundation Models with Patient-Aware Sampling** — SD4H Workshop @ ICML 2026 [arXiv]
**J. Placidi**, Y. Liu, J. Han, M. Rei, and A. A. Faisal

**Autoregressive EHR Foundation Models with Multimodal Inputs** — SD4H Workshop @ ICML 2026 [arXiv]
Y. Liu*, **J. Placidi***, J. Han*, A. J. Balston, M. Rei, and A. A. Faisal
* *Joint first authors*

**Scene Text Recognition with Semantics** — arXiv preprint 2022 [arXiv]
**J. Placidi**, Y. Miao, Z. Wang, and L. Specia

## Experience

**Senior Machine Learning Engineer** **Better Origin**
September 2022 – April 2025

- Built autonomous robotic insect farms that recycled food waste into protein.
- Led development of the software and ML infrastructure.
- Trained models to maximise protein yield across varying feed and environmental conditions.
- Scaled robot automation from 0% to 99.97%, saving 3 hours of manual work per farm daily.
- Built the company's central data platform, enabling biologists to analyse unit-level trends, refine processes, and guide ML model development.

**Founder** **London AI Group**
January 2024 – April 2025

- Founded an AI consultancy delivering agentic systems and RAG pipelines to SMEs.

**Tutor and Lab Assistant** **University of Edinburgh**
January 2022 – May 2022

- Led tutorials and labs for the Foundations of Natural Language Processing module.

**Research Intern** **LAMA Lab @ Imperial College London**
September 2020 – September 2021

- Developed scene-text recognition methods for heavily distorted and partially occluded text, supervised by Yishu Miao and Lucia Specia.

**ML Intern** **Circuit Mind**
May 2021 – September 2021

- Trained language encoders in low-data settings, improving accuracy by 31% [blog].

**Intern** **The Oxford Foundry @ University of Oxford**
July 2020 – September 2020

- Helped a postdoctoral team spin out SLAM-based 3D reconstruction into a robotics company.

## Skills

**ML:** Multimodal FMs, Transformers, ViTs, GNNs, NLP/LLMs, PyTorch, Distributed Training
**Software Engineering:** Python, Git, SQL, Cloud Infrastructure, Web Applications, Docker, Linux

## Selected Work

- A Life Is Worth $10^x$ Tokens, blog post [link], 2026.
- Attribution Graphs: How to Read an Artificial Brain, blog post [link], 2025.
- Optimising BSF insect rearing with ViTs [link] – Talk, University of Cambridge, 2023.
- Generative AI for Fungi (Fun-GAI) [link], BioHack London (2nd place), 2023.
- LLM fine-tuned to my WhatsApp, YouTube [link], 2023.